

BENIKAM SRIKAR [4pt] *Computer Science Student · Distributed Systems · Visual Computing Research Enthusiast* [6pt] ;
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EDUCATION

B.Tech in Computer Science and Engineering

Sasi Institute of Technology and Engineering

Relevant Coursework: Data Structures, Operating Systems, Computer Networks, DBMS, OOP

Aug 2023 – May 2027

CGPA: 8.45 / 10

RESEARCH INTERESTS

Distributed Systems, Computer Graphics, Computer Vision, Machine Learning, Visual Computing, Real-Time Systems, Human-Computer Interaction, AI-assisted Content Creation

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Python, C++
Backend	Node.js, Express, REST APIs, Socket.IO, WebRTC
Frontend	React, Next.js, Tailwind CSS
AI / ML	PyTorch, TensorFlow
Tools & Infra	Docker, AWS, Supabase, Git, Linux

PROJECTS

SWIFT (Secure Wideband Instant File Transfer)

Solo Project / Live Demo / GitHub

Real-time file transfer platform with zero persistent storage, live upload-download model, and integrated real-time chat

- Architected a zero-storage file transfer system where transfers happen live between sender and receiver simultaneously – unlike WhatsApp, Gmail, or Telegram which store files server-side until the receiver comes online; only cryptographic transfer logs are retained as proof-of-transfer
- Improved transfer reliability by ~40% and reduced connection setup time by ~30% through iterative architecture redesign across versions
- Designed a signaling backend using Socket.IO to manage concurrent sessions, peer discovery, and real-time event coordination
- **v1.0:** Implemented core WebRTC-based single-file transfer using room ID-based peer discovery
- **v1.2:** Added multi-file transfer with ZIP-based chunking for large payloads (>25MB) and integrated Google OAuth 2.0 authentication
- **v1.3:** Replaced room-based discovery with email-based host matching, reducing user onboarding friction and simplifying connection flow
- **v1.4:** Integrated Google Drive fallback for large file transfers (>25MB), increasing successful delivery rate by ~60%
- **v1.5:** Re-architected transfer model from peer-to-peer WebRTC to a live upload-download pipeline – sender uploads in real time while the receiver downloads simultaneously with zero server-side file storage; removed Google Drive dependency entirely
- **v1.6:** Implemented real-time chat supporting both group and individual messaging within rooms, enabling text and link sharing alongside file transfers without any third-party messaging dependency

Research and System Design [-6pt]

VEDA — Visual Enhancement and Denoising Assistant

Research-Oriented Project / In Progress

AI-assisted post-processing framework for Blender that enhances Cycles-rendered frames using computer vision and deep learning for professional cinematic quality

- Identified noise and grain artifacts in complex volumetric (fog) and dynamic (explosive) scenes from personal Blender experimentation since intermediate days; inspired by advanced VFX techniques such as DNEG's physics-based rendering for Interstellar, developed a practical asynchronous AI post-processing pipeline without modifying Blender's core engine
- Architected a decoupled dual-component architecture (lightweight Blender addon + independent VEDACore background service) to separate rendering from GPU-accelerated AI workloads

- Designed an automated render-event ingestion pipeline featuring file-system monitoring, real-time frame detection, asynchronous transfer, denoising/enhancement, and output to a dedicated enhanced directory
- Implemented a cinematic formatting subsystem supporting professional standards (Cinemascope 35/70mm, IMAX 35/70mm) through automatic resolution and aspect-ratio adaptation
- Integrated and evaluated state-of-the-art open-source models — NAFNet for neural denoising and Real-ESRGAN for super-resolution & restoration — to significantly reduce noise while preserving geometric details and artistic intent
- Explored critical systems challenges in process orchestration, inter-process communication, asynchronous workflow management, and GPU inference optimization
- Demonstrates research potential in Computer Graphics, Computer Vision, Deep Learning-based Image Restoration, Operating Systems, and AI-assisted Digital Content Creation

CERTIFICATIONS

OCI Generative AI Professional	Oracle Cloud Infrastructure	2024
OCI AI Foundations	Oracle Cloud Infrastructure	2024
Backend Development & Databases	Coursera	2024
Linux Shell Scripting	Coursera	2024

ACTIVITIES & INTERESTS

- Built and experimented with real-time systems concepts through hackathon environments, focusing on low-latency communication and event-driven architectures
- Participated in Google and AMD Hack2Skill hackathons, contributing to prototype systems involving real-time interaction and AI-assisted features
- Explored internal workings of WebRTC, signaling mechanisms, peer-to-peer networking, and live streaming pipelines through implementation-driven learning
- Interested in distributed systems design, scalable backend architectures, and production-grade real-time communication systems
- Actively building intuition around system design trade-offs through projects involving WebSockets, WebRTC, and backend orchestration